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The Chihuahua vs. Muffin workshop sort of introduced, the fundamentals of image classification through a playful but still meaningful machine learning exercise: you train a model, to tell images of chihuahuas from images of muffins. At first glance the two groups might seem unrelated, but visually they can start to look similar because of overlapping textures, colors, and overall shapes. The main point of the session was really to show how computer vision models can be trained to notice those subtle visual differences between two image classes. And while doing that, the workshop also touched on key machine learning ideas like image preprocessing, convolutional neural networks (CNNs) , and a topic sometimes described as transfer learning.

One of the most important things I picked up from the workshop is that image classification isn't just about "looking" at an image the way humans do. A machine learning model has to learn patterns from the pixel data, and pull out useful features that let it tell one class from another. CNNs are really good for this job, because they're made to work on images by picking up hierarchical patterns, for example edges shapes textures and other stuff that becomes more involved in the later layers. In the lab I got a less fuzzy view of how CNNs are put together, and why they're so commonly used in computer vision. Instead of me manually defining features, CNNs essentially learn them on their own from the training data which is why they're so strong for image recognition type tasks (LeCun, Bengio, & Hinton, 2015).

In the workshop one of the big ideas I kept coming back to was transfer learning, and honestly I found it pretty valuable. The main point is that you don't have to train a deep learning model entirely from scratch every single time. Instead, transfer learning lets you begin with a model that is already been trained on a huge image dataset, like ImageNet, and then you adapt it to a brand new classification task. Here the workshop basically treated a pre trained model like a starting block for classifying chihuahuas and muffins. What that ends up doing is making training a lot faster and more efficient, because the earlier layers are already tuned to general visual things like edges, textures, and object forms or shapes. TensorFlow also notes that this kind of transfer learning is very useful when the available dataset is rather small, since it cuts down on the need for massive amounts of labeled examples, while still keeping performance strong. Overall, this workshop helped me grasp why transfer learning is such a practical strategy in real life machine learning projects.

It was engaging in the lab, but also it had these problems that showed up. Like, figuring out how the model actually gets better with training, not just at the end but gradually. At first I zoomed in on the final accuracy, and I didn't always think about what was going on in each training epoch, and that was my mistake. I had to pay more attention to how training accuracy and validation accuracy relate, and also to the risk of overfitting. If the model looks great on the training set but then falls apart on the validation set, it feels like it is memorizing the examples instead of really learning patterns that can generalize. To push through this, I went through the notebook step by step, matched each chunk of code to the concept behind it, and spent more time on how preprocessing, model architecture, and the training decisions ended up shaping the final outcome.

The approaches from this workshop have a lot of practical use cases in the real world. Such as image classification is often everywhere in healthcare, where models can support spotting illnesses from medical scans; in security systems too, where facial recognition or maybe object detection gets used; in retail, where items can get sorted automatically and more or less consistently; and with autonomous vehicles, where computer vision is basically key for figuring out pedestrians, traffic signs, and those road conditions. Even the funny “chihuahua vs. muffin” example mirrors an actual problem in machine learning, meaning it’s about separating categories that look pretty similar at first glance. So, the whole exercise feels more than just a playful thing, it really shows how image classifiers need to grasp those small differences, the ones that can matter a great deal in professional environments.

Overall this workshop felt like a valuable learning experience, it mixed the theory with hands on practice in a way that works. I also really liked how the lab made ideas like CNNs, and transfer learning seem a lot more approachable, almost like they were less abstract. They did it through a creative and kind of memorable example, which was nice. On my side it was challenging but also motivating. It showed me that machine learning isn’t only about applying models, it’s also about understanding how they learn from data, where they can fail or wobble, and how you can improve them over time. As I keep studying artificial intelligence, this workshop gave me a sturdier foundation in computer vision and I guess a better appreciation for the practical role deep learning plays when you’re trying to solve real classification problems.

Citation

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.

TensorFlow. (n.d.). *Transfer learning and fine-tuning*. TensorFlow Tutorials.

https://www.tensorflow.org/tutorials/images/transfer_learning